

ForeDance: Structured Streaming Dance Generation with Anticipatory Music Conditioning

Abstract—Streaming dance must prepare for music that has not yet arrived while continuing from motion that can no longer be revised. We present ForeDance, which uses a frozen music foundation model to forecast future music states from arrived audio and condition robot-native dance generation. Kinematics-guided discrete-continuous representation and paired reconstruction objectives give whole-body structure and local detail complementary roles. Commit Forcing trains continuation from their jointly generated prefix and the reference state induced by its decoded movement. The framework repeatedly plans and commits motion, with a reference interface to a fixed whole-body controller. On 198 generated 60-second FineDance trajectories, a frozen music-motion evaluator favors the intended soundtrack over alternative gallery tracks in 74.20% and 75.01% of pairwise comparisons under two startup settings, with positive margins across all generator training seeds. These results support soundtrack correspondence in the generated references. ForeDance connects musical foresight, structured motion generation, and learning to continue from committed movement within a causal streaming framework.

I. INTRODUCTION

Dance preparation begins before a visible accent: weight shifts precede a step, and the body transitions toward a pose before reaching it. A system dancing to music as it arrives must choose these movements before hearing all the sound they will accompany. Waiting for an event leaves less time to prepare; reading the future recording changes the task into offline generation. Streaming further requires each new movement to continue from an output prefix that can no longer be revised. Musical anticipation and motion continuation are therefore coupled requirements of streaming dance generation.

Music-conditioned choreography has advanced through expressive motion models and large paired datasets [1]–[4]. Streaming systems such as DiscoForcing [5] encode incoming audio causally, supporting motion generation as the soundtrack unfolds. Our question is how to supply prospective musical context when future audio is unavailable. We investigate this question in robot-native dance: the model produces reference motion in the humanoid’s own configuration space, so musical choices must be expressed through its body and continued across successive planning decisions.

We present ForeDance, a framework that conditions streaming dance on predicted future music states. A frozen Magenta RealTime 2 (MRT-2) music foundation model [6] rolls forward from the audio received so far. This transfers a general musical continuation prior into dance generation: paired dance data teaches the motion model to use the prior, while musical prediction is learned through large-scale music pretraining. To our knowledge, ForeDance is the first

streaming dance generation framework to explicitly condition motion on future music states forecast by a pretrained music foundation model using past and present audio alone. The predicted states guide both whole-body organization and complementary articulation.

Realizing this design requires a representation that organizes movement and a training procedure that accounts for earlier generated choices. Our kinematics-guided discrete-continuous (D+C) representation couples discrete structure and continuous detail with complementary reconstruction objectives. Structure-only reconstruction preserves designated body organization and endpoint placement; their combination reconstructs full motion through a shared decoder. Building on hybrid motion representations [7], this supervision specifies what the two components should preserve. Commit Forcing then trains continuation after jointly sampling and committing both components. Their decoded movement determines the next starting state, pairing generated history with the reference state it actually induces.

At each decision, ForeDance plans a short segment, commits its prefix, and updates the context for the next plan (Fig. 1). A reference interface delivers the resulting stream to a fixed SONIC whole-body controller [8]. The generator continues from committed references; measured robot feedback is handled by the controller. This keeps the motion-generation mechanism explicit while providing a path to motion tracking.

Our contributions are:

- **Anticipatory music conditioning for streaming dance.** Predicted future states from a frozen music foundation model provide musical lookahead using arrived audio alone.
- **Kinematics-guided D+C representation and paired training.** Complementary reconstruction objectives establish structural and detail roles for coupled music-conditioned generation through a shared decoder.
- **Commit Forcing for streaming continuation.** Jointly generated structure and detail define both the committed history and its resulting reference state for continuation training.

We evaluate soundtrack correspondence in 198 generated 60s FineDance trajectories. A frozen G1-native music-motion evaluator favors the intended soundtrack over alternative gallery tracks in 74.20% and 75.01% of pairwise comparisons under two startup settings, with positive margins in every training seed. Complementary motion-quality measurements and rendered references characterize the generated

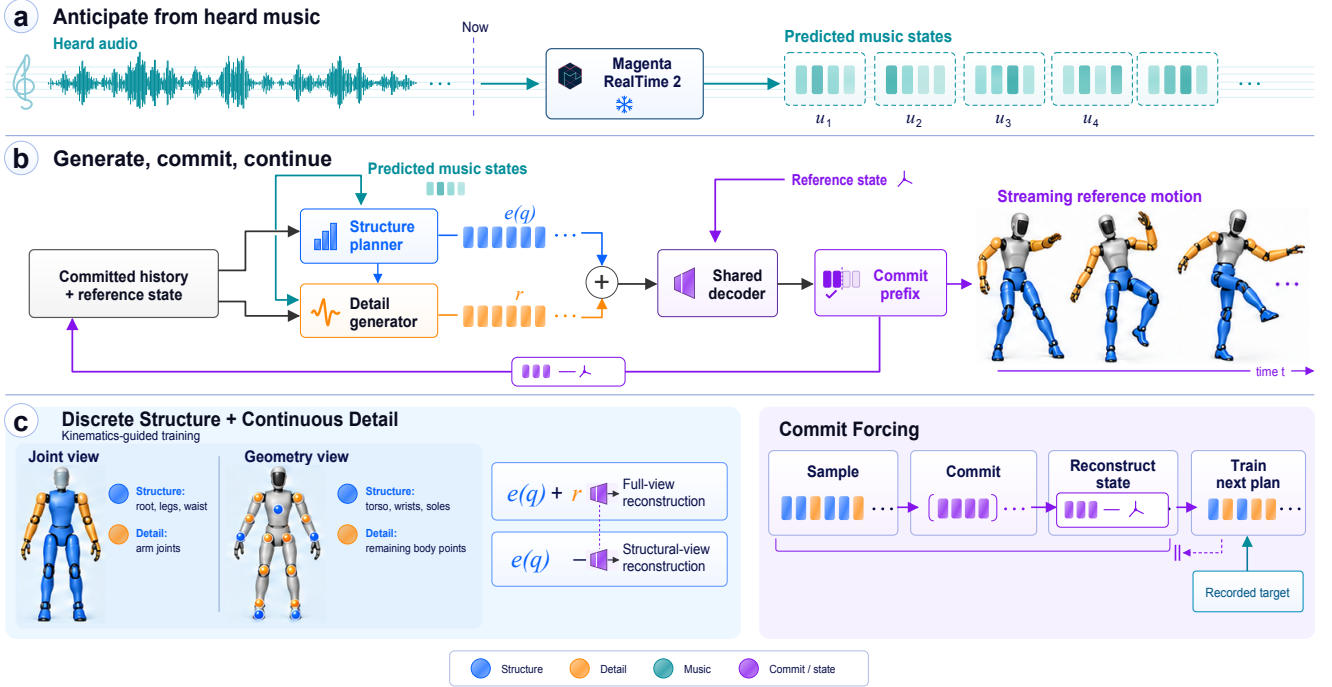


Fig. 1. **ForeDance overview.** **Top:** heard audio supplies a causal forecast of upcoming music states. **Middle:** the forecast conditions both discrete structure and continuous detail; structure also conditions detail. Their sum passes through a shared decoder, and the committed prefix updates the motion history and reference-derived boundary state for the next decision. **Bottom:** kinematic supervision gives the two latent components complementary roles, while Commit Forcing trains continuation from a jointly generated history–state pair. Robot poses illustrate the representation and output interface; they are schematic, not experimental or hardware execution evidence.

dance. These completed studies establish correspondence within the framework; Section V distinguishes this evidence from the comparisons needed to attribute forecast and component benefits.

II. RELATED WORK

A. Observed and Predicted Music Conditions

Music-to-dance models combine acoustic descriptors [9] or learned features [10], [11] with choreography models. FACT and Bailando model coordinated sequences and choreographic memory [1], [12]; EDGE and LODGE develop diffusion-based and long-form generation [2], [13]. FineDance, MEGADance, and SoulNet address fine-grained motion, genre, and music–motion alignment [3], [4], [14].

DiscoForcing [5] uses a task-specific causal VQ-PAE encoder to describe heard audio through rhythmic tokens and periodic alignment. We instead condition on explicit future states from a generative music prior. MRT-2 is pretrained on approximately 71,000 hours of music [6], providing continuation knowledge beyond our paired dance training. Foundation-model features already support dance in EDGE through Jukebox [2], [15] and in DGFM [16]; earlier robot systems also predict beat timing [17]. Our distinction is the use of generative music-state forecasts under a causal audio constraint. Live Music Models [18] and MRT-2 supply the music-generation foundation; we transfer its predictions to structured streaming motion.

B. Structure–Detail Representation and Training

Discrete tokens support motion generation in T2M-GPT and MoMask [19]–[21]; MLD uses continuous latent diffusion [22], and DisCoRD uses flow-based decoding from discrete tokens [23]. Most closely, DC-Motion [7] combines discrete structure with continuous residuals. We build on this hybrid principle, using designated robot-kinematic supervision to give the components complementary roles through a shared, boundary-conditioned decoder. Neural Categorical Priors and GPC learn vocabularies for physical control [24], [25]; our latents instead decode into explicit motion references for a separate tracker.

C. Continuation from Generated Context

Scheduled Sampling and Professor Forcing address training–sampling mismatch [26], [27]. Diffusion Forcing varies per-token noise [28]; MotionStream and MotionStreamer develop causal motion generation [29], [30]. MotionStreamer’s Two-Forward strategy uses first-pass predictions in second-pass history. Commit Forcing constructs the joint D+C history–state transition: sampled components enter history together, and their decoded prefix determines the next reference state. Its distinction is this coupled update. Diffusion Policy and Real-Time Chunking address committed actions during plan updates [31], [32]; CoF constructs the context under which continuation is learned.

a) Robot execution: Reusable trackers support expressive motion [8], [33]–[35], while HARMON and RoboPerfom introduce language and audio conditioning [36], [37]. PhysDiff, terrain-aware reference generation, and PARC incorporate physical correction or adaptation [38]–[40]. Our fixed-controller interface keeps reference generation separate from measured-state feedback, allowing their outputs to be examined independently.

III. METHOD

ForeDance links musical forecasting, structured motion generation, and continuation after a committed prefix. Figure 1 shows the common forecast condition and the motion context shared by these operations.

a) Streaming task: At a planning decision, $a_{\leq \tau}$ is the arrived audio, h is committed motion-code history, and b is the pose and velocity state derived from the last committed reference frames. A frozen music model produces future states $m^{\text{pred}} = F(a_{\leq \tau})$. The generator samples

$$(q_{1:H}, r_{1:H}) \sim p_{\theta}(q, r \mid h, b, m^{\text{pred}}), \quad (1)$$

where q and r are discrete structure and continuous detail. A shared decoder maps them to native G1 motion. We commit the first $C < H$ latent steps, discard the suffix, and update history and the reference state. Thus later decisions may revise uncommitted plans while preserving the emitted motion.

A. Anticipatory Music Conditioning

The next movement will accompany sound that is only partly observed. We provide prospective context by tokenizing arrived audio with SpectroStream and rolling frozen MRT-2 Small forward [6]. Its general music pretraining supplies the continuation prior; paired dance training learns how to translate the predicted states into movement. The forecast covers approximately 0.56 s, alongside a motion plan of approximately 0.53 s. No future waveform samples enter the condition.

Each forecast state carries its intended time. Learned projections and time embeddings align these states to motion positions, and frame-wise feature modulation [41] conditions both branches:

$$\text{FiLM}(v_i, c_i) = \gamma(c_i) \odot v_i + \beta(c_i). \quad (2)$$

Here v_i is a generator activation, c_i its aligned music condition, and γ, β predict a scale and shift. Conditioning both branches lets the same prospective context guide body organization and local articulation; structure also conditions the detail generator. The music model stays frozen while these motion-side mappings are learned. Expired forecast states are masked by their timestamps. Appendix I specifies timing and conditioning details.

B. Kinematics-Guided D+C and Paired Training

Musically directed motion needs coordinated body organization together with local articulation. We learn these

roles through complementary supervision, using a discrete–continuous latent

$$z_i = e(q_i) + r_i, \quad (3)$$

where $e(q_i)$ is a codebook embedding and r_i a continuous residual. A shared decoder $D_{\psi}(z, b)$ maps their sum and starting state to robot-native motion. Quantization supplies a structural hypothesis; the residual adjusts movement within it. The training objectives determine which properties this division should preserve (Fig. 2).

a) Learning complementary roles: We pair full-latent reconstruction $D_{\psi}(e + r, b)$ with structure-only reconstruction $D_{\psi}(e, b)$. The full path reconstructs all motion views, whereas the structure-only path is supervised on designated structural views. Both use the same decoder and starting state. This gives discrete structure an explicit kinematic responsibility while allowing the combined latent to retain complete movement.

The supervision views cover native coordinates and velocities, plus forward-kinematics positions and velocities. In joint space, structure selects root, legs, and waist; detail contains the arm coordinates. In geometry, structure selects the torso, both wrists, and both sole points; detail contains the remaining body points. Thus wrist placement can be structural while arm-joint articulation belongs to the complementary native view. These are supervision views of shared decoding, not separate anatomical decoders. An auxiliary structural loss under residual corruption reinforces the same organization when detail varies. Exact partitions, reconstruction gradients, and loss weights are given in Appendix I.

b) Coupled motion generation: A causal Transformer samples structural tokens. A residual diffusion generator [42] conditions on that sampled structure, the common history and starting state, and the forecast music. Their sum decodes into 38-dimensional G1 motion, including root movement and 29 joint angles. Both branches commit the same prefix, preserving their pairing across plans. The next segment starts from a reference-derived pose and velocity state; the coordinate convention and complete state definition are in Appendix I.

C. Commit Forcing

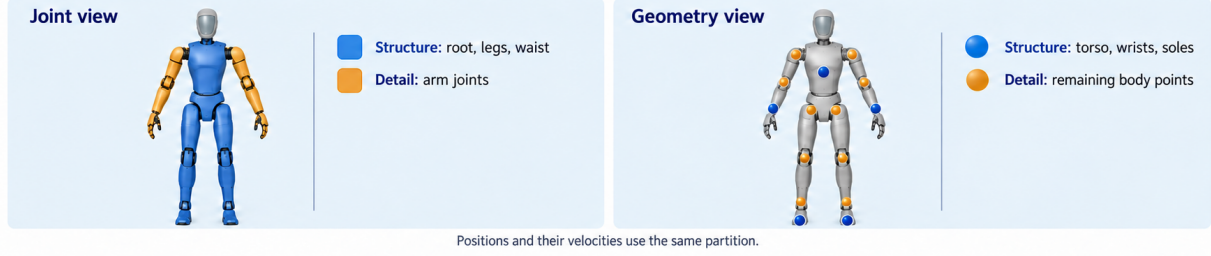
Every committed segment changes both latent history and the reference state from which the next segment begins. Teacher Forcing (TF) supplies these inputs from recorded motion; streaming inference obtains them from generated motion. Commit Forcing constructs this generated transition before supervising continuation (Fig. 3). A stop-gradient first pass samples structure $\tilde{q}$ and detail $\tilde{r}$ with the inference samplers. Their common committed prefix is decoded, and an update operator U constructs

$$(\tilde{h}^+, \tilde{b}^+) = U(h, b, \tilde{q}_{1:C}, \tilde{r}_{1:C}). \quad (4)$$

The next training pass receives the appended codes together with the state induced by their movement. This keeps history and state consistent with the same generated commitment rather than combining unrelated transitions.

Kinematics-guided structure-detail decoupling

(a) Body-part supervision views



(b) Two principal reconstruction paths

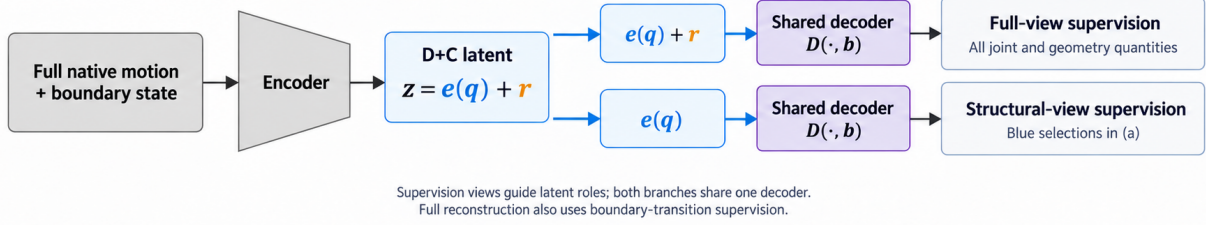


Fig. 2. **The representation and its training are one design.** Body-part supervision has separate joint and geometry partitions: root/legs/waist versus arms in joint space, and torso/wrists/soles versus other body points in geometry. The two principal reconstruction paths supervise the full latent on all views and discrete structure on the structural view. The shared decoder and boundary state are used throughout.

a) *Learning the next plan:* The next recorded latent remains the training target. For target $z^* = e(q^*) + r^*$ and sampled next structure $\hat{q}$, we rebase the residual:

$$r_{\text{rebased}}^* = z^* - e(\hat{q}). \quad (5)$$

Consequently, $e(\hat{q}) + r_{\text{rebased}}^* = z^*$: rebasing preserves the target latent. Decoded movement also depends on the starting state and need not be identical under recorded and generated states. Each example independently constructs its first-pass transition; the procedure trains continuation after a generated commit without requiring a fully recursive training rollout. The distinguishing operation is the joint history–state update, which exposes the learner to the coupled consequences of its structure and detail predictions.

b) *Supporting history regularization:* Full-history corruption (FHC) perturbs discrete embeddings and residuals across the motion history, after CoF context construction. It leaves the reference state and music condition unchanged and is disabled at inference. Appendix I gives its settings.

D. Robot Reference Integration and Execution

Figure 4 separates reference delivery from controller feedback. The committed native motion provides the reference interface to a separate, pretrained SONIC whole-body controller [8]. A reference bridge reconstructs root motion in the fixed sequence coordinate system, maps joint references into the controller convention, and resamples the 30 Hz motion onto a continuous 50 Hz grid. Appendix I defines the coordinates and state reconstruction. Only committed motion is delivered; the uncommitted suffix is discarded before the next plan and is not sent to the controller.

SONIC remains fixed and combines the consumed reference with measured robot state to produce control actions.

The generator separately updates its latent history and boundary state from committed reference motion. Appendix IV specifies the delivery and measurement protocol.

IV. EXPERIMENTS

Our evaluation distinguishes musical correspondence, motion quality, and execution. The completed studies characterize forecast-conditioned reference motion and supporting tracking behavior. Section V identifies the comparisons still needed to attribute the core method’s gains.

A. Setup and Evaluation

The correspondence study contains 198 generated 60 s FineDance trajectories in native G1 space: eleven tracks, two startup settings, three independently trained generators and three sampling seeds per generator. We call the settings *prefix-aware startup* and *direct-prefix startup*; the study characterizes both without selecting a winner. All associated tracks form a fixed music gallery, with their original audio offsets retained. A separate motion-quality study uses the full archived 18-track cohort. Training settings and the distinct cohort protocols are specified in Appendices I and II.

a) *Music–motion correspondence:* Motivated by music–motion retrieval [4], we use a frozen G1-native evaluator trained independently of the generator. Its score $s_w(x, a)$ averages cosine similarities from three evaluator instances for five-second music–motion windows. For intended music a_i and another gallery track a_j , we compute

$$\delta_{ijw} = s_w(x_i, a_i) - s_w(x_i, a_j). \quad (6)$$

Pairwise correspondence accuracy counts positive margins, with half credit for ties; random ordering gives 50%. This

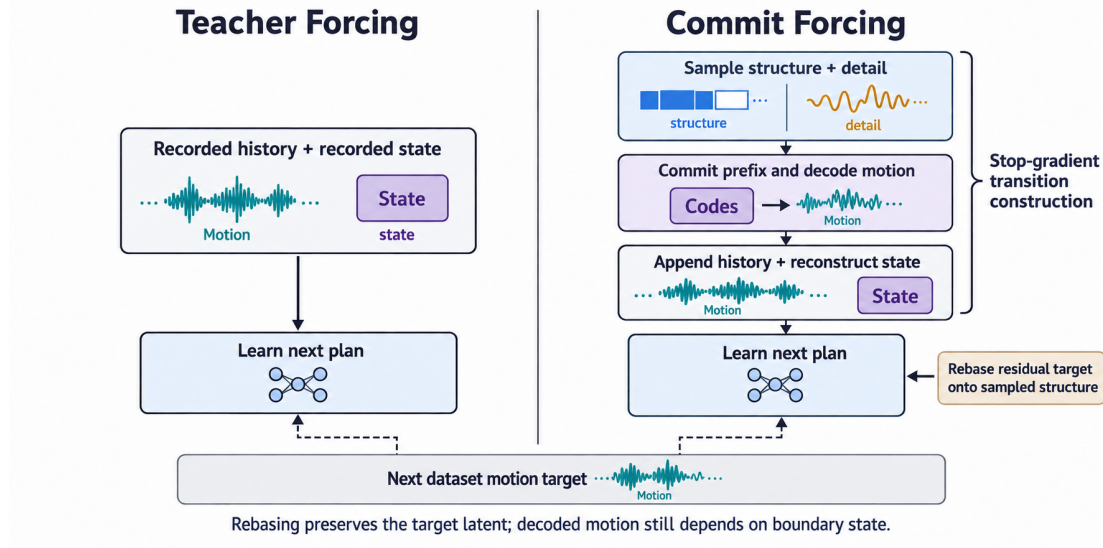


Fig. 3. **Learning the next plan: TF versus CoF.** TF receives a recorded history–state pair. CoF samples and commits structure and detail together, derives the corresponding next reference state, and supervises continuation. Both use recorded targets. Residual rebasing retains the target latent under sampled structure; decoding additionally depends on the supplied boundary state.

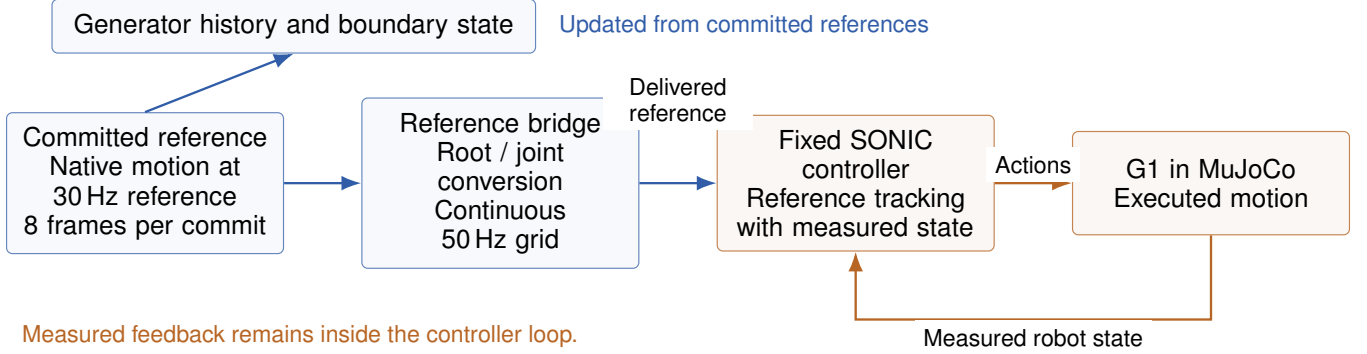


Fig. 4. **From committed references to motion tracking.** The bridge converts the reference stream onto the controller grid. Committed references update the generator context, whereas measured robot state closes the SONIC tracking loop.

measures preference for the intended soundtrack while motion is fixed, rather than eleven-way retrieval accuracy or overall dance quality. All ten alternative tracks, windows and sampling repeats contribute. Descriptive 95% intervals resample training seeds and query tracks with the gallery fixed; overlapping windows are not treated as independent samples. Appendix II-A gives the evaluator and its calibration, and Appendix II gives aggregation details.

b) Motion quality: Kinetic/geometric Fréchet distances and diversity characterize generated motion distributions. Root/contact proxies and jerk complement these measures with grounding and smoothness information. The 18-track results are analyzed within their own cohort rather than pooled with soundtrack correspondence.

B. Soundtrack Correspondence and Generated Motion

Table I shows consistent preference for intended music. Correct-pair cosine exceeds wrong-track cosine by 0.1786 and 0.1783 for prefix-aware and direct-prefix startup, respec-

tively. Correct music wins 74.20% and 75.01% of pairwise comparisons. Every generator training seed has a positive mean margin in both settings, supporting correspondence across independently trained models in this gallery.

a) Complementary motion quality: The 18-track study examines FHC as a supporting training choice. Distribution distances improve most under mixed-start training, while contact and jerk favor clean history. The full values in Appendix III document this distribution–smoothness tradeoff alongside the correspondence results.

b) Qualitative generation: Figure 5 shows reference poses at 10, 30, and 50 s from one selected 60 s rollout. These original rendered frames illustrate body configuration across the sequence; they do not measure temporal smoothness.

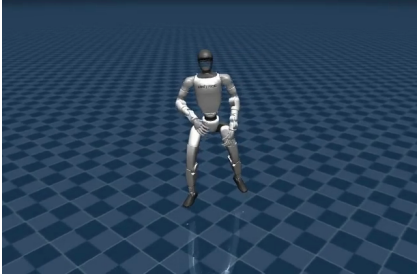
C. Supporting Execution Diagnostics

The interface in Section III-D separates the generated, resampled, controller-consumed and measured trajectories. The supporting study replays four saved trajectories from

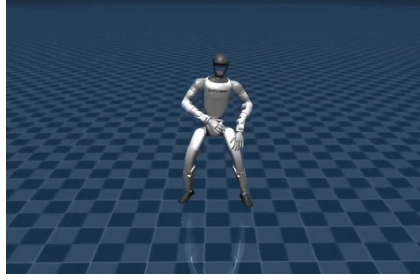
TABLE I

SOUNDTRACK CORRESPONDENCE OF 60-SECOND GENERATED REFERENCES. EACH FIXED MOTION IS SCORED WITH ITS INTENDED MUSIC AND TEN OTHER TRACKS IN THE SAME ELEVEN-TRACK GALLERY. COSINE AND CORRECT-MINUS-WRONG MARGIN ARE HIGHER-IS-BETTER; PAIRWISE ACCURACY COUNTS CORRECT-MUSIC WINS WITH HALF CREDIT FOR TIES. INTERVALS ARE DESCRIPTIVE 95% BOOTSTRAP INTERVALS WITH THE GALLERY FIXED.

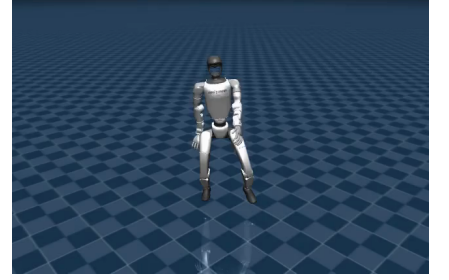
Startup	Correct	Wrong	Margin	Margin interval	Pairwise accuracy (%) [interval]
Prefix-aware startup	0.2105	0.0319	+0.1786	[0.0938, 0.2633]	74.20 [63.77, 84.15]
Direct-prefix startup	0.2057	0.0274	+0.1783	[0.0983, 0.2608]	75.01 [65.17, 84.13]



10 seconds



30 seconds



50 seconds

ForeDance: selected 60-second FineDance generation.

Fig. 5. **A 60-second ForeDance generation.** Original frames at 10, 30, and 50 s from a selected FineDance sequence. The images show rendered reference motion.

one FineDance track through fixed SONIC in MuJoCo. These trajectories use an earlier 34-dimensional representation, separate from the current 38-dimensional generation cohort; audio inference is not run during replay.

After excluding the first two seconds and aligning joint trajectories, median amplitude retention ranges from 0.858 to 0.901, and median squared-velocity retention from 0.411 to 0.496. Position tracking therefore coexists with dynamic attenuation. These diagnostics characterize the reference-to-execution path; they do not establish current-model hardware performance or comparative musical benefit. Appendix IV retains the complete measurements and protocol.

V. DISCUSSION

a) Supported findings and attribution: The completed 198-trajectory study supports soundtrack correspondence within the fixed gallery. Isolating the benefit of anticipation requires matched forecast-versus-heard conditioning; attributing representation and continuation gains requires independently trained D-only/C-only/D+C comparisons and CoF-versus-TF rollouts. These comparisons remain pending. Broader music pretraining supplies a transferable prediction prior, while improved out-of-distribution dance generalization remains a hypothesis. The historical replay diagnostics characterize tracking separately from these current reference-generation results.

b) Evaluation scope and selection: All available gallery tracks and training/sampling repeats are retained. The evaluator configuration was selected after inspection of a 30-content-group calibration cohort; its calibration scores and fixed-gallery intervals are descriptive, not selection-independent evidence of configuration superiority. The score

has not been validated against human preference or executed trajectories. The motion-quality cohort is separate, and the illustrated sequence is a selected qualitative case. Its complete source and selection record are retained with the evidence.

VI. CONCLUSION

ForeDance brings anticipatory music conditioning to structured streaming dance: a frozen music foundation model predicts prospective context, kinematics-guided D+C training organizes motion components, and Commit Forcing learns continuation after their joint commitment. Completed evaluation finds consistent soundtrack correspondence across generator training seeds. Together, the method and reference interface provide an explicit framework for studying how musical foresight translates into continuously generated robot motion.

APPENDIX I

REPRESENTATION AND TRAINING DETAILS

a) Native representation and boundary state: We represent motion directly in the robot's configuration space, including full pelvis orientation. A fixed yaw anchor R_A is removed once for a motion sequence, and every frame stores absolute orientation in that anchor:

$$R_t^A = R_A^\top R_t^{\text{world}}, \quad (7)$$

$$x_t = [\Delta p_{xy,t}^{\text{local}}, h_t, \rho(R_t^A), \theta_t]. \quad (8)$$

Here $\Delta p_{xy,t}^{\text{local}}$ denotes planar displacement, h_t is pelvis height, ρ is the six-dimensional rotation representation [43], and θ_t contains 29 joint angles, giving 38 motion dimensions. This retains roll and pitch without multiplying a sequence of predicted rotation increments. Planar displacements are

integrated using the appropriate preceding heading. The anchor remains fixed over the generated sequence; it is not reset at every commit.

The corresponding 71-dimensional boundary state is

$$b_t = [h_t, v_{xy,t}^{\text{local}}, v_{z,t}, \rho(R_t^A), \omega_t, \theta_t, \dot{\theta}_t]. \quad (9)$$

The state contains pelvis height, local planar and vertical velocities, orientation, angular velocity ω_t , joint angles, and joint velocities. Conditioning on these quantities makes the starting pose and motion explicit at each segment boundary. They are reconstructed from the committed reference.

b) Streaming configuration: The motion rate is 30 Hz, with two frames per latent step. We use 64 history steps (128 frames, approximately 4.27 s), an eight-step plan (16 frames, approximately 0.53 s), and a four-step commit (eight frames, approximately 0.27 s). The codec latent has 16 dimensions and its codebook has 512 entries. The music condition uses 14 native forecast states at 25 Hz, spanning approximately 0.56 s. Each state retains its physical timestamp.

c) Reconstruction gradients and auxiliary objectives:

Let sg denote stop-gradient. For encoder output z , embedding e , and $r = z - \text{sg}(e)$, we form full and structure-only reconstructions, together with an auxiliary residual-corrupted reconstruction:

$$x_f = D_\psi(\text{sg}(e) + r, b), \quad (10)$$

$$x_s = D_\psi(z + \text{sg}(e - z), b), \quad (11)$$

$$x_c = D_\psi(\text{sg}(e) + \mathcal{C}(r), b). \quad (12)$$

The full branch is supervised on Φ_f ; the other two are supervised on Φ_s . The corruption $\mathcal{C}$ either zeros detail or adds small channel-scaled noise, encouraging the structural reconstruction to persist under changes in the residual. The combined objective is

$$\mathcal{L}_{\text{codec}} = \mathcal{L}_f + \mathcal{L}_s + 0.25\mathcal{L}_{\text{robust}} + \mathcal{L}_{\text{VQ}} + \mathcal{L}_{\text{boundary}}. \quad (13)$$

The final term supervises boundary-transition derivatives and pelvis orientation at the beginning of the reconstructed segment. Appendix I provides the training details.

d) Conditioning and sampling: For forecast state u_j with lead ℓ_j relative to the motion decision, we form $c_j = W \text{LN}(u_j) + \phi(\ell_j)$, where W is a learned projection, LN denotes layer normalization, and ϕ represents physical time. Timestamp-based masking excludes expired forecast states from their aligned motion positions. A null condition bypasses music modulation. The structural planner uses a classification objective and the detail model a denoising objective. Residual generation uses ten DDIM steps [44].

e) Kinematic reconstruction losses: The full view contains four blocks: native motion, native velocity, forward-kinematics positions, and geometry velocity. Each block is standardized with training-split statistics. For a view Φ with blocks $\Phi^{(k)}$, define the reconstruction error

$$\mathcal{E}_\Phi(\hat{x}, x) = \frac{1}{4} \sum_{k=1}^4 \text{mean} \left[\left(\Phi^{(k)}(\hat{x}) - \Phi^{(k)}(x) \right)^2 \right]. \quad (14)$$

This gives $\mathcal{L}_f = \mathcal{E}_{\Phi_f}(x_f, x)$, $\mathcal{L}_s = \mathcal{E}_{\Phi_s}(x_s, x)$, and $\mathcal{L}_{\text{robust}} = \mathcal{E}_{\Phi_s}(x_c, x)$. Equal block weights prevent the number of geometric coordinates from determining their relative contribution. The VQ term is the codebook loss plus 0.25 times the commitment loss.

For the auxiliary robustness objective, the residual is zeroed with probability 0.5. Otherwise, Gaussian noise is added with scale sampled uniformly from 0.01–0.05 times the residual channel standard deviation. The resulting reconstruction is supervised on the structural view. This codec objective is separate from FHC, which regularizes the generator’s motion history.

The boundary loss supervises the first three decoded frames relative to the supplied boundary. It combines normalized smooth- L_1 errors for velocity, acceleration, and jerk with weights 1, 0.5, and 0.25, respectively, plus pelvis-rotation geodesic error with weight 1. These quantities explicitly penalize a reconstructed segment whose beginning is inconsistent with the target transition.

f) Body-part supervision views: In the native block, structure selects the first 24 of 38 coordinates: nine root coordinates, twelve leg joints, and three waist joints. The remaining 14 coordinates are the bilateral shoulder (three per arm), elbow (one), and wrist (three) joints. In the 35-dimensional velocity block, structure selects the first 21 coordinates: root linear/angular velocity and the same 15 leg/waist joint velocities. The other 14 are arm-joint velocities.

The geometry block contains 31 points: 29 non-root body origins and two lowest-foot geometry points. Structure selects the torso, the two wrist-yaw body origins, and the two lowest-foot points. Detail selects the remaining 26 points. Geometry velocity uses the same partition. These are supervision and analysis views, not independent anatomical decoders. In particular, the structural geometry view can constrain wrist placement while the complementary native view describes arm articulation.

g) Generator optimization and initialization: Generator training uses 6,250 updates with batch size 48, corresponding to 300,000 training windows. Mixed-start training draws 20% song-origin starts and 80% continuation windows; continuation-start training uses ordinary windows throughout. Residual sampling uses guidance fixed at 1.0; the music model remains frozen during generator training.

h) History regularization: FHC applies to the complete 64-step history with probability 0.5 per example. The discrete embeddings and continuous residual history share a severity timestep sampled from 850–999, with independent Gaussian noise. Their masks and timestamps are preserved, as are the boundary state and forecast conditions. Corruption is applied in the supervised pass after CoF context construction and is disabled at inference.

i) Constructing CoF transitions: Figure 6 details the construction. Each training example independently samples a first-pass transition from its supplied context. The committed discrete and residual prefixes enter the history together, and their decoded motion determines the next boundary

state. The next recorded structural and latent targets remain fixed. Residual rebasing expresses that latent target relative to the sampled structure; it does not re-encode the target under the generated boundary. This construction trains local continuation after a generated commit without requiring a fully recursive long training rollout.

APPENDIX II EVALUATION PROTOCOL

The primary correspondence cohort yields 99 trajectories per startup variant: three generator training seeds, three sampling seeds, and eleven tracks. Each trajectory contains 1,800 frames at 30 Hz and supplies 56 overlapping five-second windows with one-second stride. For every motion, the intended audio is compared with all ten wrong-track donors at matched relative windows, preserving each source’s native audio offset. Means are formed over windows and donors, then sampling repeats within training-seed/query-track cells, and finally the three training seeds and 11 query tracks. Each bootstrap draw resamples three training seeds and eleven query tracks with replacement; the same sampled query-track list is applied across the selected training seeds. Percentiles 2.5 and 97.5 over 10,000 draws define descriptive 95% intervals. The gallery of donors remains fixed. The separate 18-track motion-quality cohort uses common source-supported 23 s segments beginning at frame 264, three training seeds and three sampling seeds, yielding 162 outputs per configuration. Its scores and uncertainty are kept separate from the 60 s correspondence study.

A. Music–Motion Evaluator

a) Frozen representation and readout: The G1-native evaluator receives paired five-second windows: 150 audio feature frames at 30 Hz and 100 motion feature frames at 20 Hz. Motion features are extracted once from each complete trajectory, then normalized with frozen training-split statistics before windowing. The continuous 357-dimensional frontend retains full root rotation in $SO(3)$; it does not reset the orientation anchor separately for every scoring window. The two encoders produce deterministic posterior means for each window in FP32 inference. For evaluator instance k , denote these embeddings by $\mu_w^{a,k}$ and $\mu_w^{x,k}$. The adopted score is

$$s_w(x, a) = \frac{1}{3} \sum_{k=1}^3 \frac{\langle \mu_w^{x,k}, \mu_w^{a,k} \rangle}{\|\mu_w^{x,k}\|_2 \|\mu_w^{a,k}\|_2}. \quad (15)$$

Scores are averaged across the three independently trained evaluator instances before pairwise ranking; embeddings and final accuracies are not averaged instead. The checkpoints use standard content-group sampling and the final training update (14,500) for each seed, with no best-seed choice. Encoder weights and normalization remain fixed across both generator startup variants. An absolute score difference of at most 10^{-7} is treated as a tie and receives half credit. This specifies the correspondence statistic in Eq. (6) without conflating it with retrieval recall.

b) Evaluator calibration: On reference motion from a 30-content-group calibration cohort, the adopted score obtains 68.41% ordinary wrong-music pairwise accuracy and 64.08% with confusing music donors. Each task contains 15 disjoint song pairs; the cohort contains 3,661 windows. These are evaluator-characterization measurements, separate from the generated-motion gallery. Multiple evaluator configurations were inspected on this cohort, so the values are descriptive as stated in the Discussion. The fixed standard-trained ensemble is retained for generation scoring; calibration and generated-motion scores do not establish perceptual validity on executed robot trajectories.

c) Reproducible score aggregation: Each of the eighteen generator-training/sampling combinations supplies an $11 \times 11 \times 56$ music–motion score array. Its diagonal contains intended pairs and each off-diagonal column supplies the ten donors for one fixed motion. All 198 per-motion scores, margins, and accuracies are reproduced from these arrays; no source or window is removed based on its score. The frozen cohort, evaluator version, per-track records, and complete arrays are retained with the evaluation evidence.

d) G1 distribution and local-motion features: Kinetic descriptors summarize motion dynamics, while geometric descriptors summarize body configurations and spatial relations in the G1 domain. FID compares feature means and covariances with those of retargeted references; diversity measures pairwise distances in the same feature space. Extractor version, coordinate convention, normalization, crop length, and sample counts are part of each result. The earlier three-model table and current ForeDance results are not pooled: their numerical feature-scale equivalence has not been established. A multi-seed FID aggregation must retain how collections were formed; it is not a mean of single-song FIDs. For future execution scoring, generation and execution must use identical feature extraction. Exact feature lists and activity/contact thresholds require the corresponding implementation receipt.

e) Beat and dynamic-response measures: For motion-event times $\mathcal{B}_x$ and music-beat times $\mathcal{B}_a$, the motion-to-music BAS definition is

$$\text{BAS} = \frac{1}{|\mathcal{B}_x|} \sum_{b \in \mathcal{B}_x} \exp\left(-\frac{\min_{a \in \mathcal{B}_a} (b - a)^2}{2\sigma^2}\right). \quad (16)$$

The earlier layer analyzer uses minima of smoothed mean joint speed with $\sigma = 0.10$ s. This does not automatically specify the extractor used for the current-generation BAS summaries, which retain their own metric version. Event counts, empty-event handling, smoothing, and any beat-matching tolerance must accompany reproducible scores. Beat precision/recall are auxiliary coverage diagnostics, not requirements that dance hit every audio beat. The layer analyzer’s impact proxy is a smoothed absolute derivative of the speed curve, not contact force. Zero-lag correlation with audio onset strength measures aligned dynamic response; best-lag correlation and its lag are diagnostics, not permission to retime primary music scores.

1. CONSTRUCT ONE GENERATED TRANSITION (STOP GRADIENT)

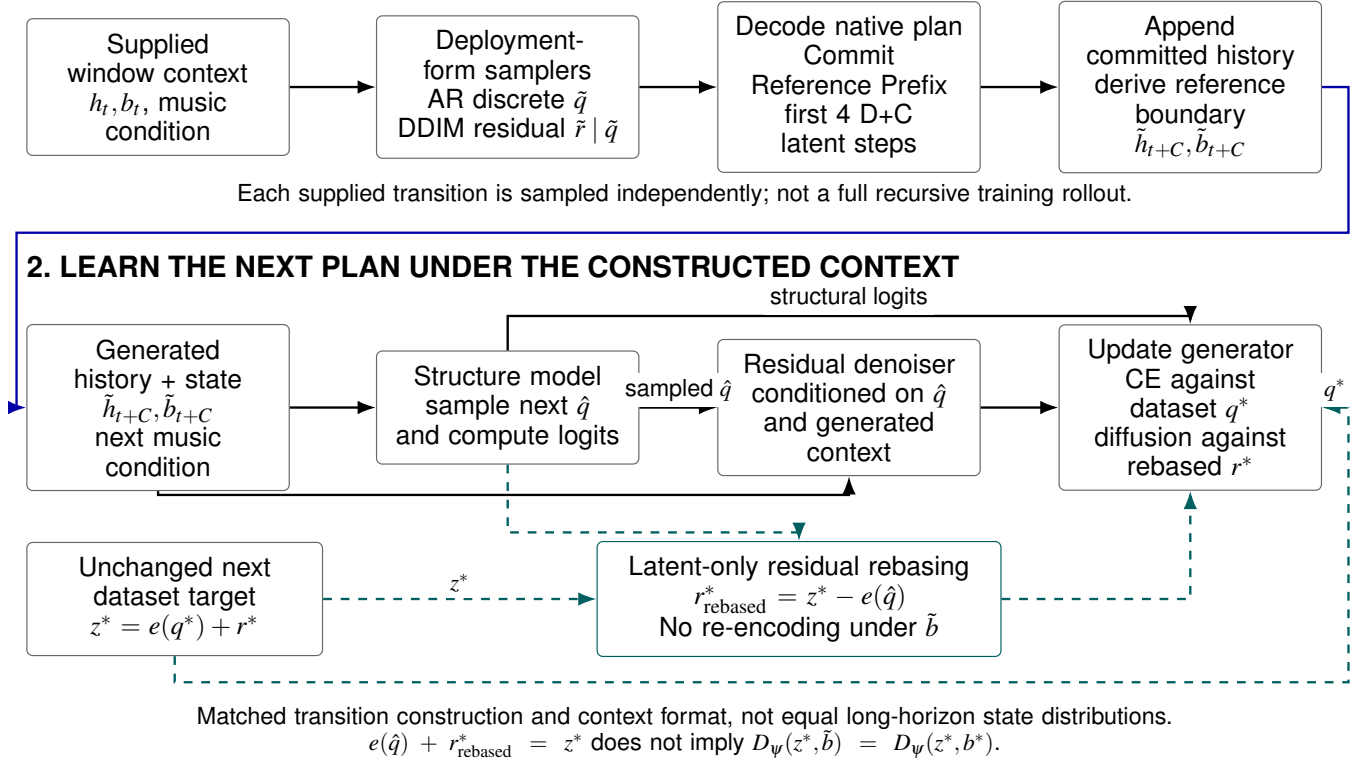


Fig. 6. **Commit Forcing transition construction and supervision.** A sampled, jointly committed structure–detail prefix supplies the next history and reference boundary. The next dataset latent remains fixed; residual rebasing expresses it relative to the sampled next structure. This preserves the latent target, while the changed boundary can change its decoded motion. The two stages are constructed independently for each supplied training example.

APPENDIX III

HISTORY-CORRUPTION RESULTS

Table II provides the full values for the comparison in Section IV-B. Under mixed-start training, kinetic/geometric FID changes from 208.599/0.482 with clean history to 162.831/0.410 with FHC, with the same favorable direction across training seeds. Continuation-start changes are smaller, from 156.218/0.372 to 154.180/0.361. Mixed-start contact and jerk favor clean history, exposing a distribution-smoothness tradeoff.

APPENDIX IV

REFERENCE DELIVERY AND SUPPORTING EXECUTION DIAGNOSTICS

a) *Replay setup:* Four generated trajectories from one FineDance track and one sampling seed are replayed in MuJoCo for approximately 60 s each.

b) Reference construction and delivery: A four-step latent commit contains eight 30Hz frames. A continuous 50Hz time grid yields 13 or 14 reference samples per commit without resetting the sampling phase at each packet. The controller receives committed references rather than motor torques. The four diagnostic replays use the same fixed SONIC controller at real-time playback, with no uncommitted-plan preview, clipping, startup ramp, or alignment hold. Initial state, delivered future-reference access, and

packet timestamps are part of the protocol; full-trajectory delivery and committed-prefix delivery are not interchangeable controls. The trajectories begin at an artificial continuation point with empty motion history, rather than at full-song onset.

c) Observed layers: Let M_{gen} , M_{sent} , M_{target} , and M_{exec} denote generated, resampled, consumed, and measured motion. Target and measured joints use named fields in MuJoCo order, with matched coordinate conventions, offsets, and timestamps.

d) Conditions and measurements: The first three trajectories share cached forecast content but use different age traces: ideal age, the earlier measured trace, and a revised measured trace. The fourth uses live forecast content with the revised trace. These labels identify the generation conditions of the saved trajectories.

For active joint j , amplitude and velocity-variation retention are

$$R_j^{\text{amp}} = \frac{P_{95}(q_j^{\text{exec}}) - P_5(q_j^{\text{exec}})}{P_{95}(q_i^{\text{target}}) - P_5(q_i^{\text{target}})}, \quad (17)$$

$$R_j^{\text{dyn}} = \frac{\mathbb{E}[(\dot{q}_j^{\text{exec}})^2]}{\mathbb{E}[(\dot{q}_i^{\text{target}})^2]}. \quad (18)$$

Only joints with a target percentile range above 0.02 rad enter the active-joint summary, and ratios with near-zero denomina-

TABLE II

EFFECT OF HISTORY CORRUPTION ON THE FULL 18-TRACK COHORT. LOWER IS FAVORABLE FOR FID AND PFC. DIVERSITY DESCRIBES MOTION SPREAD AND IS INTERPRETED TOGETHER WITH FIDELITY.

Startup	History	FID _k	FID _g	Div _k	Div _g	PFC
Mixed-start	Clean history	208.599	0.482	7.495	0.493	0.143
Mixed-start	History corruption	162.831	0.410	8.626	0.508	0.167
Continuation-start	Clean history	156.218	0.372	8.329	0.545	0.163
Continuation-start	History corruption	154.180	0.361	9.017	0.541	0.162

tors are excluded. The reported retention values are medians of joint-wise ratios. The dynamic ratio measures squared joint velocity; it is not physical energy or a whole-body energy ratio. Joint-position RMSE is evaluated after time alignment. Alignment lag is estimated within ± 0.5 s on the 50 Hz grid; the four recorded estimates are 40 ms, which is a case-specific alignment result rather than a general controller or system latency. Steady-state summaries exclude the first two seconds, so they do not establish startup behavior.

e) *Interpretation*: Amplitude retention ranges from 0.858 to 0.901, while squared-velocity retention ranges from 0.411 to 0.496. Position tracking thus coexists with dynamic attenuation. High-frequency variation can include articulation and jitter; these measurements alone do not determine the musical value of the attenuation.

REFERENCES

- [1] R. Li, S. Yang, D. A. Ross, *et al.*, “AI Choreographer: Music conditioned 3D dance generation with AIST++,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 13401–13412.
- [2] J. Tseng, R. Castellon, and C. K. Liu, “EDGE: Editable dance generation from music,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 448–458.
- [3] R. Li, J. Zhao, Y. Zhang, *et al.*, “FineDance: A fine-grained choreography dataset for 3D full body dance generation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10234–10243.
- [4] X. Li, R. Li, S. Fang, *et al.*, “Music-aligned holistic 3D dance generation via hierarchical motion modeling,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 14420–14430. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025/html/Li_Music-Aligned_Holistic_3D_Dance_Generation_via_Hierarchical_Motion_Modeling_ICCV_2025_paper.html
- [5] K. Ji, B. Qian, B. Wu, *et al.*, “DiscoForcing: A unified framework for real-time audio-driven character control with diffusion forcing,” *arXiv preprint arXiv:2605.28491*, 2026. [Online]. Available: <https://arxiv.org/abs/2605.28491>
- [6] Magenta Team, “Magenta RealTime 2: Open & local live music models,” Magenta blog, June 2026. [Online]. Available: <https://magenta.withgoogle.com/magenta-realtime-2>
- [7] H. Wang, X. Chen, J. Zhang, *et al.*, “DC-Motion: Decoupling structure and details via discrete-continuous tokens for human motion generation,” *arXiv preprint arXiv:2606.14721*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.14721>
- [8] Z. Luo, Y. Yuan, T. Wang, *et al.*, “SONIC: Supersizing motion tracking for natural humanoid whole-body control,” *Science Robotics*, vol. 11, no. 117, p. ead4592, 2026. [Online]. Available: <https://arxiv.org/abs/2511.07820>
- [9] B. McFee, C. Raffel, D. Liang, *et al.*, “librosa: Audio and music signal analysis in Python,” in *Proceedings of the 14th Python in Science Conference*, 2015, pp. 18–25. [Online]. Available: <https://proceedings.scipy.org/articles/Majora-7b98e3ed-003>
- [10] Y. Li, R. Yuan, G. Zhang, *et al.*, “MERT: Acoustic music understanding model with large-scale self-supervised training,” in *International Conference on Learning Representations*, 2024. [Online]. Available: https://proceedings.iclr.cc/paper_files/paper/2024/hash/33dffa2e3d2ab74a783d1a8c292f66d9-Abstract-Conference.html
- [11] H. Zhu, Y. Zhou, H. Chen, *et al.*, “MuQ: Self-supervised music representation learning with mel residual vector quantization,” *arXiv preprint arXiv:2501.01108*, 2025. [Online]. Available: <https://arxiv.org/abs/2501.01108>
- [12] L. Siyao, W. Yu, T. Gu, *et al.*, “Bailando: 3D dance generation by actor-critic GPT with choreographic memory,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022, pp. 11050–11059.
- [13] R. Li, Y. Zhang, Y. Zhang, *et al.*, “LODGE: A coarse to fine diffusion network for long dance generation guided by the characteristic dance primitives,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1524–1534.
- [14] K. Yang, X. Tang, Z. Peng, *et al.*, “MEGADance: Mixture-of-experts architecture for genre-aware 3D dance generation,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/102bd3c1076dfe98043b19340ac44e08-Abstract-Conference.html
- [15] P. Dhariwal, H. Jun, C. Payne, *et al.*, “Jukebox: A generative model for music,” *arXiv preprint arXiv:2005.00341*, 2020. [Online]. Available: <https://arxiv.org/abs/2005.00341>
- [16] X. Liu, Z. Feng, D. Kanojia, *et al.*, “DGFM: Full body dance generation driven by music foundation models,” *arXiv preprint arXiv:2502.20176*, 2025. [Online]. Available: <https://arxiv.org/abs/2502.20176>
- [17] N. Nakahara, K. Miyazaki, H. Sakamoto, *et al.*, “Dance motion control of a humanoid robot based on real-time tempo tracking from musical audio signals,” in *Entertainment Computing – ICEC 2009*, ser. Lecture Notes in Computer Science, vol. 5709, 2009, pp. 36–47. [Online]. Available: https://ist.ksc.kwansei.ac.jp/~nagata/data/2009_nakahara_ICEC2009.pdf
- [18] Lyria Team, A. Caillon, B. McWilliams, *et al.*, “Live music models,” *arXiv preprint arXiv:2508.04651*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.04651>
- [19] A. van den Oord, O. Vinyals, and K. Kavukcuoglu, “Neural discrete representation learning,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017. [Online]. Available: <https://proceedings.neurips.cc/paper/2017/hash/7a98af17e63a0ac09ce2e96d03992fbc-Abstract.html>
- [20] J. Zhang, Y. Zhang, X. Cun, *et al.*, “Generating human motion from textual descriptions with discrete representations,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 14730–14740.
- [21] C. Guo, Y. Mu, M. G. Javed, *et al.*, “MoMask: Generative masked modeling of 3D human motions,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 1900–1910.
- [22] X. Chen, B. Jiang, W. Liu, *et al.*, “Executing your commands via motion diffusion in latent space,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 18000–18010.
- [23] J. Cho, J. Kim, J. Kim, *et al.*, “DisCoRD: Discrete tokens to continuous motion via rectified flow decoding,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 14602–14612.
- [24] Q. Zhu, H. Zhang, M. Lan, *et al.*, “Neural categorical priors for physics-based character control,” *ACM Transactions on Graphics*, vol. 42, no. 6, 2023.
- [25] Y. Shi, Y. Jiang, C. Tessler, *et al.*, “GPC: Large-scale generative

TABLE III

SUPPORTING FIXED-REFERENCE REPLAY MEASUREMENTS. DATA FROM THE EARLIER 34D GENERATOR, SEPARATE FROM THE CURRENT 38D COHORT. EACH ROW IS ONE MUJoCo REPLAY THROUGH FIXED SONIC, EXCLUDING ITS FIRST TWO SECONDS; MUSIC INFERENCE IS NOT RUN DURING REPLAY.

Reference-generation condition	Aligned RMSE (rad)	Median R^{amp}	Median R^{dyn}
Cached forecast, ideal age	0.1421	0.869	0.496
Cached forecast, earlier age trace	0.1281	0.858	0.428
Cached forecast, revised age trace	0.1394	0.877	0.412
Live forecast, revised age trace	0.1431	0.901	0.411

- pretraining for transferable motor control,” in *ACM SIGGRAPH 2026 Conference Papers*, 2026.
- [26] S. Bengio, O. Vinyals, N. Jaitly, *et al.*, “Scheduled sampling for sequence prediction with recurrent neural networks,” in *Advances in Neural Information Processing Systems*, vol. 28, 2015, pp. 1171–1179.
- [27] A. Lamb, A. Goyal, Y. Zhang, *et al.*, “Professor forcing: A new algorithm for training recurrent networks,” in *Advances in Neural Information Processing Systems*, vol. 29, 2016, pp. 4601–4609.
- [28] B. Chen, D. Martí Monsó, Y. Du, *et al.*, “Diffusion forcing: Next-token prediction meets full-sequence diffusion,” in *Advances in Neural Information Processing Systems*, vol. 37, 2024, pp. 24 081–24 125.
- [29] B. Jiang, X. Chen, A. Zeng, *et al.*, “Causal motion tokenizer for streaming motion generation,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision Workshops*, 2025, pp. 2045–2055. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2025W/I-HFM/html/Jiang_Causal_Motion_Tokenizer_for_Streaming_Motion_Generation_ICCVW_2025_paper.html
- [30] L. Xiao, S. Lu, H. Pi, *et al.*, “MotionStreamer: Streaming motion generation via diffusion-based autoregressive model in causal latent space,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2025, pp. 10 086–10 096.
- [31] C. Chi, S. Feng, Y. Du, *et al.*, “Diffusion policy: Visuomotor policy learning via action diffusion,” in *Robotics: Science and Systems XIX*, 2023. [Online]. Available: <https://roboticsproceedings.org/rss19/p026.pdf>
- [32] K. Black, M. Y. Galliker, and S. Levine, “Real-time execution of action chunking flow policies,” in *Advances in Neural Information Processing Systems*, vol. 38, 2025. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2025/hash/300ccb2187ded4edcc07f7e76d8e553-Abstract-Conference.html
- [33] Z. Luo, J. Cao, A. Winkler, *et al.*, “Perpetual humanoid control for real-time simulated avatars,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 10 895–10 904. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/papers/Luo_Perpetual_Humanoid_Control_for_Real-time_Simulated_Avatars_ICCV_2023_paper.pdf
- [34] M. Ji, X. Peng, F. Liu, *et al.*, “ExBody2: Advanced expressive humanoid whole-body control,” *arXiv preprint arXiv:2412.13196*, 2024. [Online]. Available: <https://arxiv.org/abs/2412.13196>
- [35] Q. Liao, T. E. Truong, X. Huang, *et al.*, “BeyondMimic: From motion tracking to versatile humanoid control via guided diffusion,” *arXiv preprint arXiv:2508.08241*, 2025. [Online]. Available: <https://arxiv.org/abs/2508.08241>
- [36] Z. Jiang, Y. Xie, J. Li, *et al.*, “Harmon: Whole-body motion generation of humanoid robots from language descriptions,” in *Proceedings of the 8th Conference on Robot Learning*, ser. Proceedings of Machine Learning Research, vol. 270. PMLR, 2025, pp. 3015–3026. [Online]. Available: <https://proceedings.mlr.press/v270/jiang25b.html>
- [37] Z. Li, C. Chi, Y. Wei, *et al.*, “Do you have freestyle? expressive humanoid locomotion via audio control,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 956–965. [Online]. Available: https://openaccess.thecvf.com/content/CVPR2026/papers/Li_Do_You_Have_Freestyle_Expressive_Humanoid_Locomotion_via_Audio_Control_CVPR_2026_paper.pdf
- [38] Y. Yuan, J. Song, U. Iqbal, *et al.*, “PhysDiff: Physics-guided human motion diffusion model,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2023, pp. 16 010–16 021. [Online]. Available: https://openaccess.thecvf.com/content/ICCV2023/html/Yuan_PhysDiff_Physics-Guided_Human_Motion_Diffusion_Model_ICCV_2023_paper.html
- [39] Z. Zhang, K. Wen, M. Xu, *et al.*, “Learning whole-body humanoid locomotion via motion generation and motion tracking,” *arXiv preprint arXiv:2604.17335*, 2026. [Online]. Available: <https://arxiv.org/abs/2604.17335>
- [40] M. Xu, Y. Shi, K. Yin, *et al.*, “PARC: Physics-based augmentation with reinforcement learning for character controllers,” in *ACM SIGGRAPH 2025 Conference Papers*, 2025.
- [41] E. Perez, F. Strub, H. de Vries, *et al.*, “FiLM: Visual reasoning with a general conditioning layer,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 32, 2018, pp. 3942–3951. [Online]. Available: <https://ojs.aaai.org/index.php/AAAI/article/view/11671>
- [42] J. Ho, A. Jain, and P. Abbeel, “Denoising diffusion probabilistic models,” in *Advances in Neural Information Processing Systems*, vol. 33, 2020, pp. 6840–6851. [Online]. Available: <https://proceedings.neurips.cc/paper/2020/hash/4c5bcfec8584af0d967f1ab10179ca4b-A-abstract.html>
- [43] Y. Zhou, C. Barnes, J. Lu, *et al.*, “On the continuity of rotation representations in neural networks,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019, pp. 5745–5753. [Online]. Available: https://openaccess.thecvf.com/content/CVPR_2019/papers/Zhou_On_the_Continuity_of_Rotation_Representations_in_Neural_Networks_CVPR_2019_paper.pdf
- [44] J. Song, C. Meng, and S. Ermon, “Denoising diffusion implicit models,” in *International Conference on Learning Representations*, 2021.